LINUX LVM-BASED DISASTER RECOVERY AND BACKUP SYSTEM

PROJECT REPORT

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Platform: Red Hat Enterprise Linux 9.8
Environment: VMware Virtual Machine
Project Type: Linux System Administration / Storage / Disaster Recovery

ABSTRACT

Data is one of the most important resources in any organization and its availability is essential for continuous business operations. Accidental deletion, file corruption, hardware failure or system-related issues can result in data loss. Therefore, organizations require reliable backup and disaster recovery mechanisms.

This project, Linux LVM-Based Disaster Recovery and Backup System, was developed on Red Hat Enterprise Linux 9.8 using a VMware virtual machine. The project uses Linux Logical Volume Manager (LVM) to create flexible storage for company data and backup files. A separate Logical Volume was created for production data, another Logical Volume was created for backup storage and an LVM snapshot was created to provide a point-in-time recovery mechanism.

The project simulates a real-world disaster recovery scenario by creating companyrelated data, taking an LVM snapshot, deleting a critical configuration file and recovering the deleted file from the snapshot. In addition, a compressed TAR backup was created and extracted separately to verify that the backup could be restored successfully.

The project demonstrates practical knowledge of Linux storage management, LVM, snapshots, backup and recovery, filesystem management, troubleshooting, file permissions and disaster recovery procedures.

INTRODUCTION

In modern IT environments, data loss can affect business operations and productivity. A system administrator must therefore ensure that important data can be backed up and recovered when required.

Linux provides Logical Volume Manager (LVM) as a flexible storage management solution. LVM allows physical storage devices to be grouped together into Volume Groups from which Logical Volumes can be created. Logical Volumes can be resized and managed independently.

One important feature of LVM is the snapshot capability. An LVM snapshot captures the state of a Logical Volume at a particular point in time. This can be useful when recovering files after accidental deletion or when creating a consistent recovery point.

In this project, an LVM-based storage environment was created on RHEL 9.8. Company data was stored in a dedicated Logical Volume while a separate Logical Volume was used for backup storage. An additional disk was later added to the Volume Group because there was insufficient free space to create the required snapshot.

The project finally demonstrates two recovery approaches: (1) LVM snapshot-based recovery and (2) TAR archive-based backup and restoration.

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1.PROJECT OVERVIEW

The project is designed to provide a basic disaster recovery and backup solution using Linux LVM.

- RHEL 9.8
- VMware virtual machine
- /dev/sdb — 10 GB additional disk
- /dev/sdc — 5 GB additional disk
- company_vg — LVM Volume Group
- data_lv — 6 GB Logical Volume for company data
- backup_lv — approximately 4 GB Logical Volume for backup storage
- data_snapshot — 1 GB LVM snapshot
- /company/data — source data location
- /company/backup — backup location

Main purpose

The main purpose is to demonstrate how an administrator can create storage, store data, create a recovery point, simulate data loss, recover data, create an independent backup, and verify the backup.

Project scenario

Assume that a company has three departments: IT, Operations and Customers. Their data is stored under /company/data.

```
/company/data/
├── IT/
├── operations/
└── customers/
```

The backup is stored separately under /company/backup.

2.PROJECT OBJECTIVE

- To understand Linux storage management.
- To understand and implement LVM.
- To create Physical Volumes, a Volume Group and Logical Volumes.
- To create separate storage for company data and backups.
- To understand LVM snapshots.
- To create a point-in-time recovery mechanism.
- To simulate accidental data deletion.
- To recover deleted data using an LVM snapshot.
- To create a compressed TAR backup.
- To verify that the backup can be restored.
- To troubleshoot insufficient LVM space.
- To demonstrate practical Linux system administration skills.

3.PROBLEM STATEMENT

Organizations continuously generate important data such as configuration files, customer information, operational records and application data.

If an important file is accidentally deleted, the organization may face:

- Data loss
- Service interruption
- Loss of configuration information
- Operational delays
- Recovery costs

Therefore, a system should have a mechanism to recover important files.

Problem addressed by this project

How can a Linux administrator create flexible storage and recover important data after accidental deletion using LVM snapshots and independent backups?

The project provides a laboratory implementation of this scenario.

4.PROJECT SCOPE

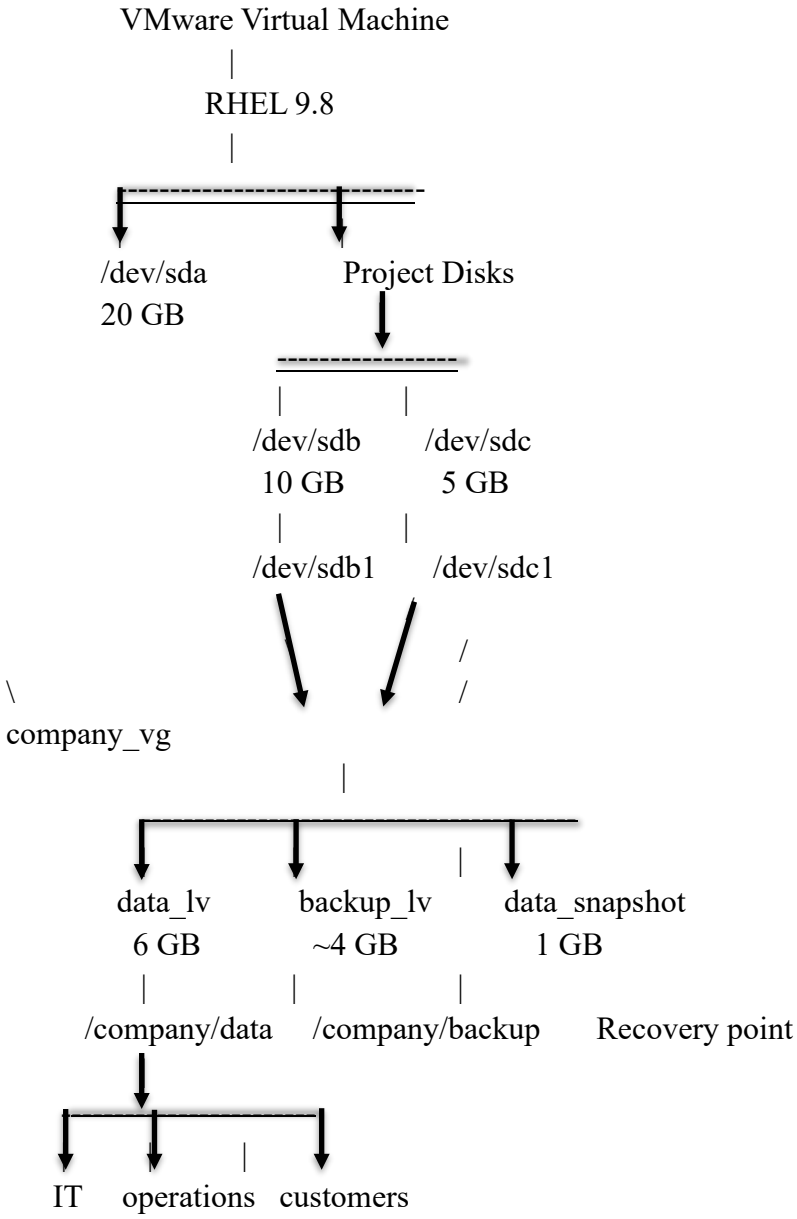
- Storage Management: Identifying disks, partitioning disks, creating LVM Physical Volumes, creating a Volume Group, creating Logical Volumes and extending the Volume Group.
- Data Management: creating company directories, creating sample business files and mounting the data Logical Volume.
- Disaster Recovery: creating an LVM snapshot, simulating accidental file deletion, mounting the snapshot and recovering the deleted file.
- Backup: creating a compressed TAR backup, listing backup contents, extracting the backup and verifying restored files.
- Troubleshooting: handling insufficient Volume Group space, adding an additional disk and extending

5.TECHNOLOGIES AND TOOLS USED

SR.NO	Technology/Tool	
1	RHEL 9.8	Operating **Purpose** system
2	VMware	Virtual machine environment
3	LVM	Storage management
4	fdisk	Disk partitioning
5	pvcreate	Create Physical Volume
6	vgcreate	Create Volume Group
7	vgextend	Extend Volume Group
8	lvcreate	Create Logical Volume/Snapshot
9	pvs	Display Physical Volumes
10	vgs	Display Volume Groups
11	lvs	Display Logical Volumes
12	lsblk	Display block devices
13	mkfs.xfs	Create XFS filesystem
14	mount	Mount filesystem
15	umount	Unmount filesystem
16	df	Check filesystem usage
17	tar	Create/extract backup archive
18	find	Locate files
19	mkdir	Create directories
20	vi	Edit files

6.SYSTEM ARCHITECTURE

Two additional virtual disks are used. /dev/sdb1 is the first LVM Physical Volume and /dev/sdc1 is added later to extend company_vg.



The operating system remains on the original /dev/sda system disk and is separate from the project storage.

7. PROJECT WORKFLOW

Start
↓
Check available disks
↓
Create partition on /dev/sdb
↓
Create Physical Volume
↓
Create company_vg
↓
Create data_lv and backup_lv
↓
Format and mount filesystems
↓
Create company data
↓
Attempt LVM snapshot
↓
Insufficient free space
↓
Add /dev/sdc
↓
Create /dev/sdc1
↓
Extend company_vg
↓
Create data_snapshot
↓
Mount snapshot
↓
Simulate accidental deletion
↓
Recover deleted file
↓
Create TAR backup
↓
Extract TAR backup
↓
Verify restored data
↓
Project Completed

8. LINUX ENVIRONMENT SETUP

The project was performed as root on RHEL 9.8. The initial block-device check showed the operating-system disk /dev/sda and an additional 10 GB disk /dev/sdb. A 5 GB disk /dev/sdc was later added to provide free LVM space for the snapshot.

8.1 Check disks lsblk

```
[root@192 ~]# lsblk
NAME                      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                         8:0    0   20G  0 disk
├─sda1                      8:1    0  600M  0 part /boot/efi
├─sda2                      8:2    0    1G  0 part /boot
└─sda3                      8:3    0 18.4G  0 part
  ├─rhel_192-root
  │                       253:0    0 16.4G  0 lvm  /
  └─rhel_192-swap
                          253:1    0    2G  0 lvm  [SWAP]
sdb                         8:16   0   10G  0 disk
└─sdb1                      8:17   0   10G  0 part
  ├─company_vg-backup_lv
  │                       253:3    0    4G  0 lvm
  └─company_vg-data_lv-real
                          253:4    0    6G  0 lvm
    ├─company_vg-data_lv
    │                     253:2    0    6G  0 lvm
    └─company_vg-data_snapshot
                          253:6    0    6G  0 lvm
sdc                         8:32   0    5G  0 disk
└─sdc1                      8:33   0    5G  0 part
  └─company_vg-data_snapshot-cow
                          253:5    0    1G  0 lvm
    └─company_vg-data_snapshot
                          253:6    0    6G  0 lvm
```

The lsblk command identifies the available disks and partitions. The 10 GB /dev/sdb disk was selected for the project.

```
[root@192 ~]#  fdisk -l /dev/sdb
Disk /dev/sdb: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x26322099

Device     Boot Start      End  Sectors Size Id Type
/dev/sdb1        2048 20971519 20969472  10G 8e Linux LVM
```

The fdisk output verifies the LVM partition /dev/sdb1.

8.2 Check operating system

cat /etc/redhat-release

hostnamectl

```
[root@192 ~]# cat /etc/redhat-release
hostnamectl
Red Hat Enterprise Linux release 9.8 (Plow)
   Static hostname: (unset)
Transient hostname: 192.168.133.132
         Icon name: computer-vm
           Chassis: vm 🖴
        Machine ID: 1bc1cbc9fb544de3b21e85c7c06ebe6f
           Boot ID: ad0fc85144274e459896c99cd5f3060e
    Virtualization: vmware
  Operating System: Red Hat Enterprise Linux 9.8 (Plow)
       CPE OS Name: cpe:/o:redhat:enterprise_linux:9::baseos
            Kernel: Linux 5.14.0-687.5.3.el9_8.x86_64
      Architecture: x86-64
   Hardware Vendor: VMware, Inc.
    Hardware Model: VMware20,1
  Firmware Version: VMW201.00V.24006586.B64.2406042154
```

These commands identify the RHEL version and provide system/hostname information.

8.3 Check IP address ip

addr

```
[root@192 ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq stat
e UP group default qlen 1000
    link/ether 00:0c:29:33:d5:04 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.133.132/24 brd 192.168.133.255 scope global dynami
c noprefixroute ens160
       valid_lft 1623sec preferred_lft 1623sec
    inet6 fe80::20c:29ff:fe33:d504/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

The project server used IP address 192.168.133.132 on the ens160 interface.

9. SOURCE DATA AND BACKUP STORAGE

9.1 Purpose of Source Data Storage

The project used separate Logical Volumes for source data and backup storage.

The Logical Volumes were:

data_lv 6 GB backup_lv

approximately 4 GB

The data Logical Volume was mounted at:

/company/data

The backup Logical Volume was mounted at:

/company/backup

Verification:

lvs

```
[root@192 ~]# lvs
  LV            VG         Attr       LSize  Pool Origin  Data%
  backup_lv     company_vg -wi-a----- <4.00g
  data_lv       company_vg owi-a-s---  6.00g
  data_snapshot company_vg swi-a-s---  1.00g      data_lv 0.20
  root          rhel_192   -wi-ao---- 16.41g
  swap          rhel_192   -wi-ao----  2.00g
```

The lvs command was used to verify the Logical Volumes created under company_vg. The data_lv Logical Volume was used for source data, while backup_lv was used for storing backup files.

9.2 Mounting the Source Data Storage

The data_lv Logical Volume was mounted at:

/company/data

This mount point acts as the main directory where the project data is stored.

The storage was verified using:

df -h

```
[root@192 ~]# df -h
Filesystem                      Size  Used Avail Use% Mounted on
devtmpfs                        804M     0  804M   0% /dev
tmpfs                           838M     0  838M   0% /dev/shm
tmpfs                           335M  7.0M  328M   3% /run
efivarfs                        256K   57K  195K  23% /sys/firmware/efi/efivars
/dev/mapper/rhel_192-root        17G  4.8G   12G  29% /
/dev/sda2                       960M  435M  526M  46% /boot
/dev/sda1                       599M  7.7M  592M   2% /boot/efi
tmpfs                           168M  104K  168M   1% /run/user/1000
/dev/sr0                        232M  232M     0 100% /run/media/viraj/CDROM
/dev/sr1                         15G   15G     0 100% /run/media/viraj/RHEL-9-8-0-BaseOS-x86_64
/dev/mapper/company_vg-data_lv  6.0G   75M  5.9G   2% /company/data
/dev/mapper/company_vg-backup_lv 4.0G  61M  3.9G   2% /company/backup
```

9.3 Creating the Source Data Directories and Server Configuration Data

mkdir -p /company/data/IT mkdir -
p /company/data/operations mkdir
-p /company/data/customers

These directories simulate different departments within a company and should contain related information.

/company
└── data
    ├── IT
    ├── operations
    └── customers/

These directories simulate different departments within a company and should contain related information.

```
[root@192 ~]# find /company/data -type f
/company/data/IT/server_config.txt
/company/data/operations/operations_data.txt
/company/data/customers/customer_data.txt
```

9.4 Create server configuration file vi
/company/data/IT/server_config.txt

The file was populated with the following information:

Server Name: RHEL-192
Operating System: Red Hat Enterprise Linux 9.8
Environment: Production
Server Role: Application Server
IP Address: 192.168.133.132
Hostname: rhel192.company. local
Backup Policy: Daily
Recovery Priority: Critical
Configuration Status: Active
Last Updated: September 2026

This file was used as a critical file for the disaster recovery demonstration.

9.4 Backup Storage

df -h



9.5 Creating the Backup Archive

After creating and verifying the source data, the /company/data directory was backed up using the TAR utility. TAR was used to combine the complete directory structure and its files into a single compressed backup archive. The backup was stored in the dedicated /company/backup directory. tar -czvf /company/backup/company_data_backup.tar.gz /company/data

```
[root@192 ~]# tar -czvf /company/backup/company_data_backup.tar.gz /company/data
tar: Removing leading `/' from member names
/company/data/
/company/data/IT/
/company/data/IT/server_config.txt
/company/data/operations/
/company/data/operations/operations_data.txt
/company/data/customers/
/company/data/customers/customer_data.txt
```

9.6 Backup Verification

```
[root@192 ~]# ls -lh /company/backup
total 4.0K
-rw-r--r--. 1 root root 689 Sep 24 12:53 company_data_backup.tar.gz
[root@192 ~]# tar -tzvf /company/backup/company_data_backup.tar.gz
drwxr-xr-x root/root         0 2026-09-21 19:12 company/data/
drwxr-xr-x root/root         0 2026-09-21 19:20 company/data/IT/
-rw-r--r-- root/root       292 2026-09-21 19:20 company/data/IT/server_config.txt
drwxr-xr-x root/root         0 2026-09-21 19:22 company/data/operations/
-rw-r--r-- root/root       297 2026-09-21 19:22 company/data/operations/operations_data.txt
drwxr-xr-x root/root         0 2026-09-21 19:22 company/data/customers/
-rw-r--r-- root/root       292 2026-09-21 19:22 company/data/customers/customer_data.txt
```

10. BACKUP SECURITY AND FILE PERMISSIONS

Linux provides ownership and permissions for protecting files. The backup and data files were created under the root account in the Linux laboratory environment.

ls -ld /company/data

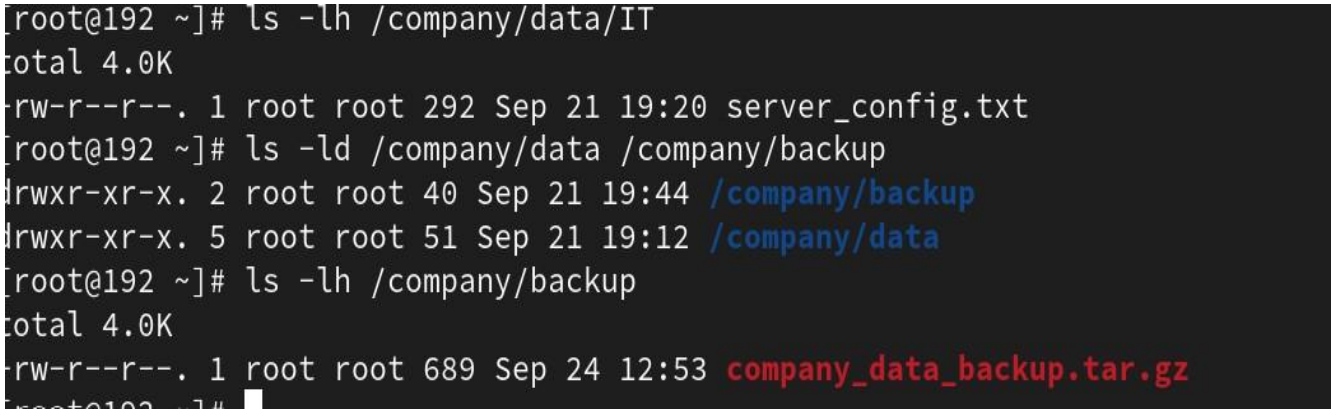
/company/backup ls -lh

/company/data/IT ls -lh

/company/backup

The backup file was owned by root. In a production environment, backup files should be protected from unauthorized access because they may contain sensitive business information.

- Root privileges were used for storage administration.
- Backup directories should have restricted permissions.
- Production backups should be stored separately.
- Sensitive backup data should preferably be encrypted.



11. BACKUP VERIFICATION

The TAR backup was created using:

tar -czvf /company/backup/company_data_backup.tar.gz /company/data

Explanation of options:

- -c = create a new archive.
- -z = compress the archive using gzip.
- -v = display files while processing.
- -f = specify the archive filename.

The resulting file was /company/backup/company_data_backup.tar.gz.

The archive was then verified: tar -tzvf

/company/backup/company_data_backup.tar.gz

```
[root@192 ~]# tar -tzvf /company/backup/company_data_backup.tar.gz
drwxr-xr-x root/root         0 2026-09-21 20:06 company/data/
drwxr-xr-x root/root         0 2026-09-21 20:05 company/data/IT/
-rw-r--r-- root/root       292 2026-09-21 20:05 company/data/IT/server_config.txt
drwxr-xr-x root/root         0 2026-09-21 20:06 company/data/operations/
-rw-r--r-- root/root       297 2026-09-21 20:06 company/data/operations/operations_data.txt
drwxr-xr-x root/root         0 2026-09-21 20:06 company/data/customers/
-rw-r--r-- root/root       292 2026-09-21 20:06 company/data/customers/customer_data.txt
[root@192 ~]#
```

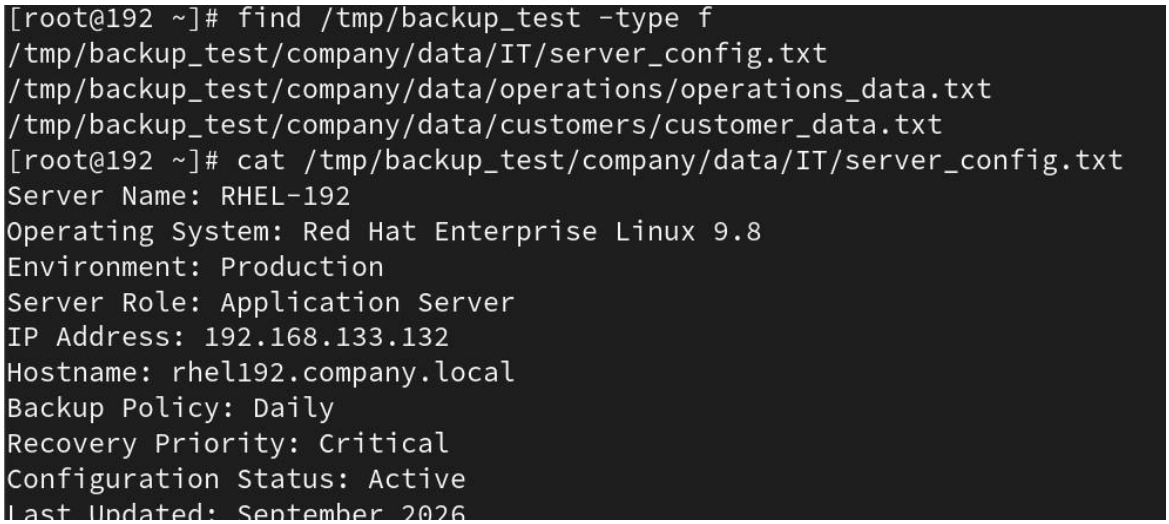
This lists the contents of the archive without extracting it.

The archive was extracted to a temporary test directory:

mkdir -p /tmp/backup_test
tar -xzvf /company/backup/company_data_backup.tar.gz -C /tmp/backup_test

The extracted files were checked:

find /tmp/backup_test -type f cat
/tmp/backup_test/company/data/IT/server_config.txt



This proves that the backup contains the required data and can be restored successfully.

12. PROJECT TESTING AND TEST CASES

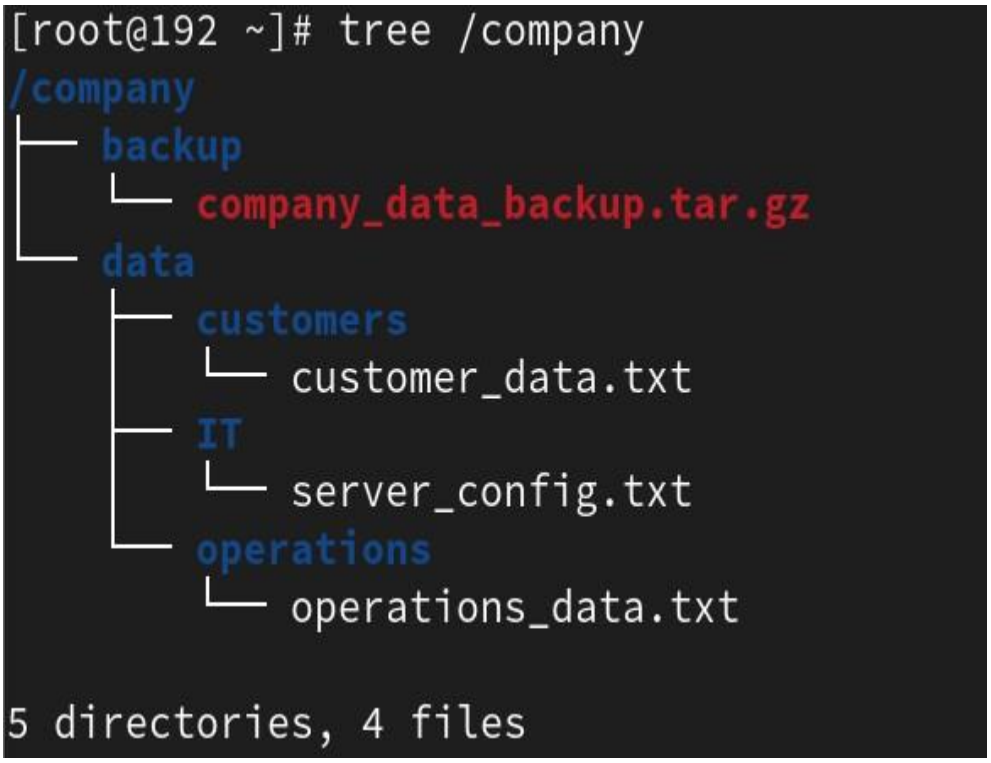
Test	Expected Result	Actual Result
Detect project disk	/dev/sdb detected	Passed
Create PV	/dev/sdb1 becomes PV	Passed
Create VG	company_vg created	Passed
Create data LV	data_lv created	Passed
Create backup LV	backup_lv created	Passed
Create snapshot initially	Fails due to insufficient space	Observed
Add second disk	/dev/sdc1 created	Passed
Extend VG	VG has free space	Passed
Create snapshot	data_snapshot created	Passed
Simulate deletion	File removed	Passed
Recover file	File restored	Passed
Create TAR	Archive created	Passed
Extract TAR	Files restored	Passed
Verify backup	Files readable	Passed

13. EXPECTED OUTPUT

The ls -lh /company/backup command should display the backup archive company_data_backup.tar.gz along with its file permissions, ownership, group, file size, and modification date. The backup file should be owned by the root user because the backup was created using root privileges.

The ls -lh /company/data/IT command should display the server_config.txt file along with its permissions, ownership, group, file size, and modification date. The file should be accessible to the root user and should exist inside the /company/data/IT directory.

The output verifies that the backup and source-data files exist and that their Linux ownership and permissions can be checked by the administrator.



14. TROUBLESHOOTING GUIDE

14.1 Insufficient Free Space During Snapshot Creation

The first snapshot attempt failed because company_vg had no free extents. The original data_lv and backup_lv consumed the available space.

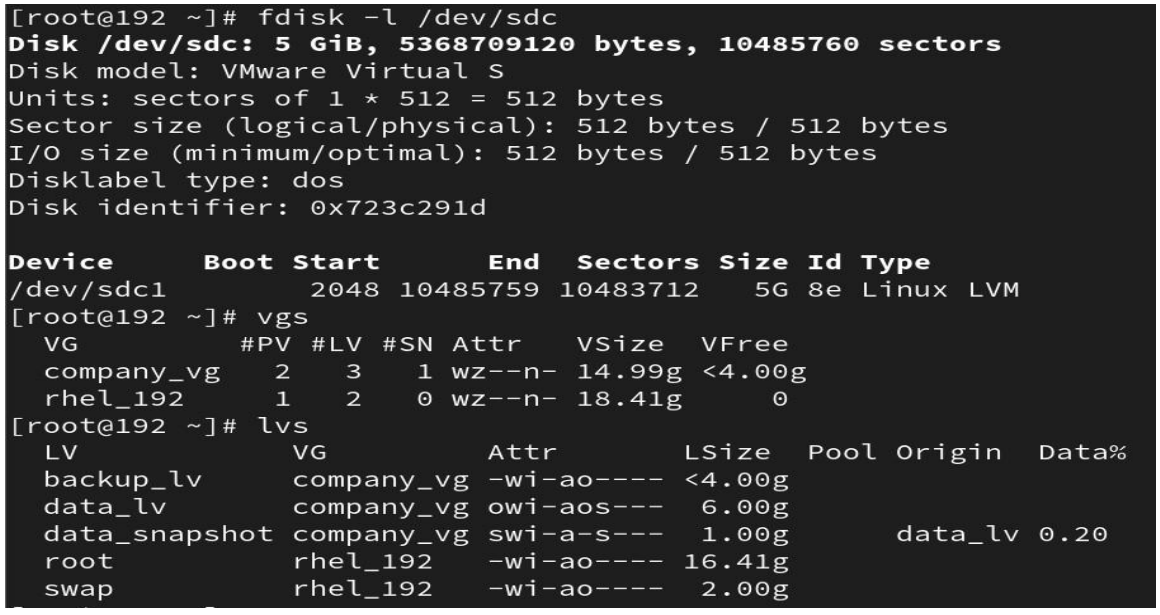
lvcreate -L 1G -s -n data_snapshot /dev/company_vg/data_lv

The command returned an insufficient free space error. Initially, company_vg was approximately 10 GB. data_lv used 6 GB and backup_lv used approximately 4 GB, leaving no free space for the 1 GB snapshot.

Solution: a second 5 GB disk was added.

pvcreate /dev/sdc1 vgextend
company_vg /dev/sdc1 vgs

After the VG was extended, approximately 5 GB became free and the 1 GB snapshot was successfully created. lvcreate -L 1G -s -n data_snapshot /dev/company_vg/data_lv



14.2 Snapshot Mount Consideration

Because an LVM snapshot of an XFS filesystem has the same filesystem UUID as the original, the snapshot was mounted with nouuid for the recovery test.

1. **Create the recovery mount point** mkdir -p /mnt/recovery
   ls -ld /mnt/recovery
2. **Mount the snapshot** mount -o nouuid /dev/company_vg/data_snapshot /mnt/recovery
3. **Verify the mount** findmnt /mnt/recovery
4. **Check the recovered files** find /mnt/recovery -type f

```
[root@192 ~]# mkdir -p /mnt/recovery
[root@192 ~]# ls -ld /mnt/recovery
drwxr-xr-x. 2 root root 6 Sep 24 13:10 /mnt/recovery
[root@192 ~]# mount -o nouuid /dev/company_vg/data_snapshot /mnt/recovery
[root@192 ~]# findmnt /mnt/recovery
TARGET        SOURCE                                  FSTYPE OPTIONS
/mnt/recovery /dev/mapper/company_vg-data_snapshot xfs    rw,relati
[root@192 ~]# find /mnt/recovery -type f
/mnt/recovery/IT/server_config.txt
/mnt/recovery/operations/operations_data.txt
/mnt/recovery/customers/customer_data.txt
```

15. SECURITY CONSIDERATIONS

- Use appropriate administrative privileges for storage-management commands.
- Restrict access to backup and recovery directories.
- Protect backup files from unauthorized modification or deletion.
- Do not keep the only production backup on the same storage system.
- Verify backup contents regularly.
- Consider encryption for sensitive production backups.
- Use appropriate access controls for administrators and application users.

16. ADVANTAGES

- Flexible storage management through LVM.
- Logical Volumes can be managed independently.
- Volume Groups can be extended by adding Physical Volumes.
- LVM snapshots provide point-in-time recovery.
- Independent TAR backup provides a second recovery method.
- The project demonstrates actual recovery and troubleshooting.

17. LIMITATIONS

- An LVM snapshot is not a substitute for an independent backup.
- Snapshots consume storage space from the Volume Group.
- A snapshot can become unusable if its allocated copy-on-write space is exhausted.
- The TAR backup in this lab is stored on the same Linux environment.
- Backup creation is manual.
- No remote replication, automated retention or centralized monitoring was implemented.

18. FUTURE ENHANCEMENTS

- Automate backups using Cron or a systemd timer.
- Implement backup retention and automatic cleanup.
- Copy backups to a separate host or remote storage.
- Encrypt sensitive backup archives.
- Add checksum-based integrity verification.
- Implement monitoring and alerting.
- Schedule snapshot creation and retention.

19. SKILLS DEMONSTRATED

- RHEL Linux administration.
- Disk and partition management.
- LVM Physical Volume, Volume Group and Logical Volume administration.
- Filesystem creation and mounting.
- LVM snapshot creation and recovery. • Linux file and directory management.
- TAR backup and restore.
- Backup verification.
- Troubleshooting insufficient LVM space.
- Command-line administration and system verification.
- Basic security and permissions awareness.

20. PROJECT CONCLUSION

The Linux LVM-Based Disaster Recovery and Backup System was successfully implemented on Red Hat Enterprise Linux 9.8 using a VMware virtual machine. The project demonstrated the complete process of creating LVM storage, configuring Logical Volumes, storing simulated company data, creating an LVM snapshot and performing data recovery.

During implementation, a snapshot creation error occurred because the Volume Group had no free space. This issue was resolved by adding a 5 GB virtual disk, creating a new LVM Physical Volume and extending the existing Volume Group. After this expansion, the required LVM snapshot was successfully created.

A simulated disaster was then performed by deleting a critical server configuration file. The file was successfully recovered from the LVM snapshot, demonstrating the effectiveness of point-in-time recovery. In addition, a compressed TAR backup was created and independently extracted and verified.

The project provided practical experience in Linux storage administration, LVM, backup and recovery, troubleshooting, filesystem management and disaster recovery. The system can be further enhanced by implementing automated scheduling, remote backup, encryption, retention policies and monitoring.